

Translated

From: e-maxx.ru

Randomized Heap

A randomized heap is a heap that, through using randomization, allows to perform all operations in expected logarithmic time.

A **min heap** is a binary tree in which the value of each vertex is less than or equal to the values of its children. Thus the minimum of the tree is always in the root vertex.

A max heap can be defined in a similar way: by replacing less with greater.

The default operations of a heap are:

- Adding a value
- Extracting the minimum
- Removing the minimum
- Merging two heaps (without deleting duplicates)
- Removing an arbitrary element (if its position in the tree is known)

A randomized heap can perform all these operations in expected $O(\log n)$ time with a very simple implementation.

Data structure

We can immediately describe the structure of the binary heap:

```
struct Tree {  
    int value;  
    Tree * l = nullptr;  
    Tree * r = nullptr;  
};
```

In the vertex we store a value. In addition we have pointers to the left and right children, which are point to null if the corresponding child doesn't exist.